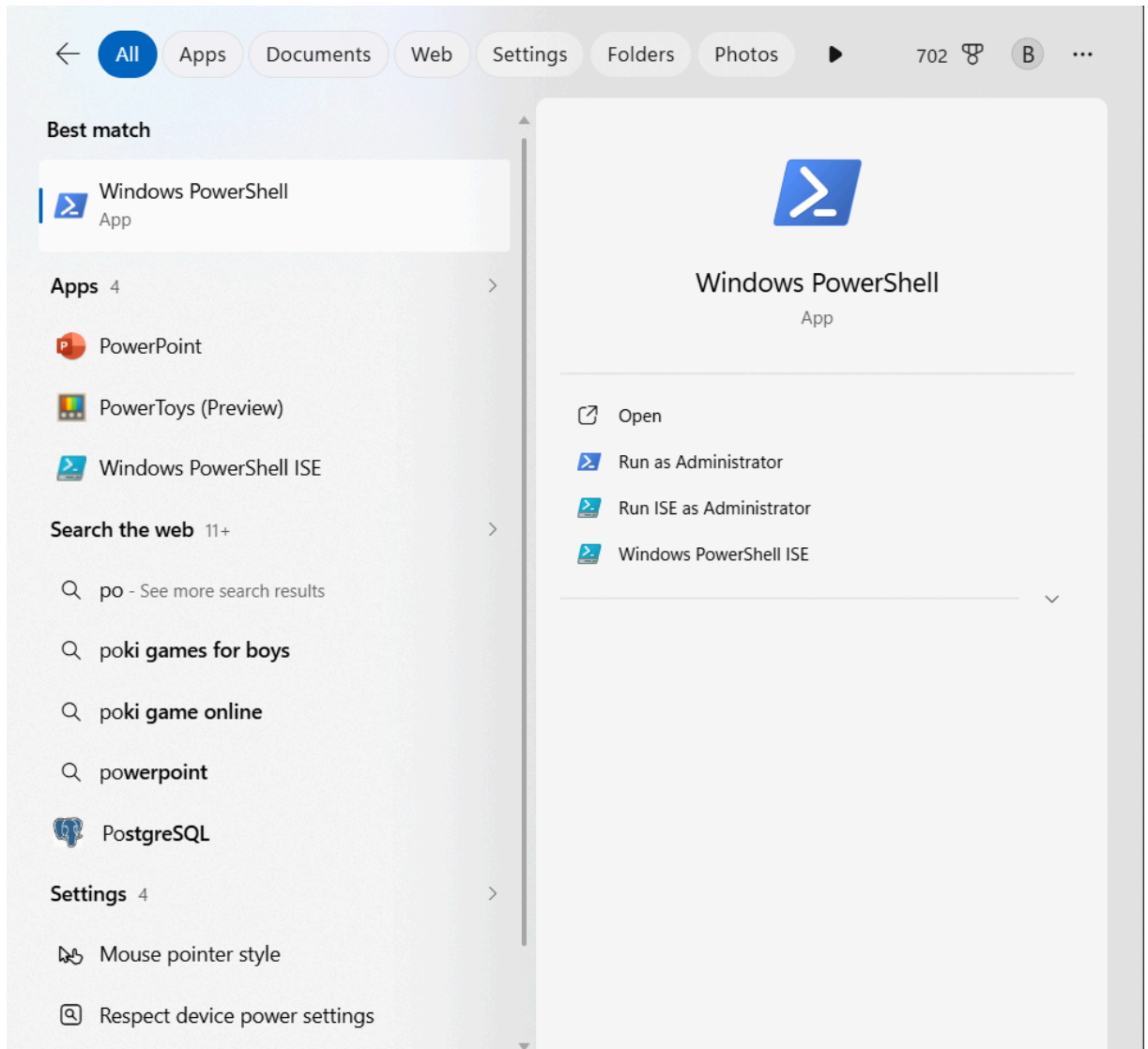
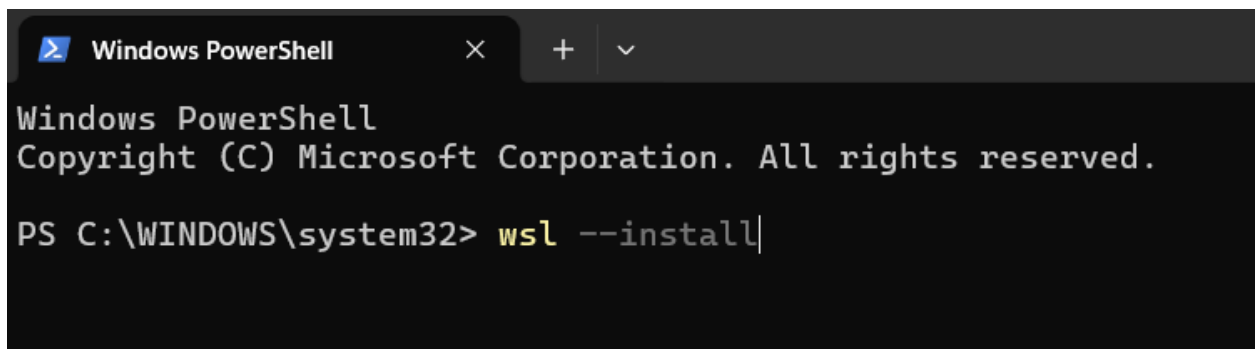
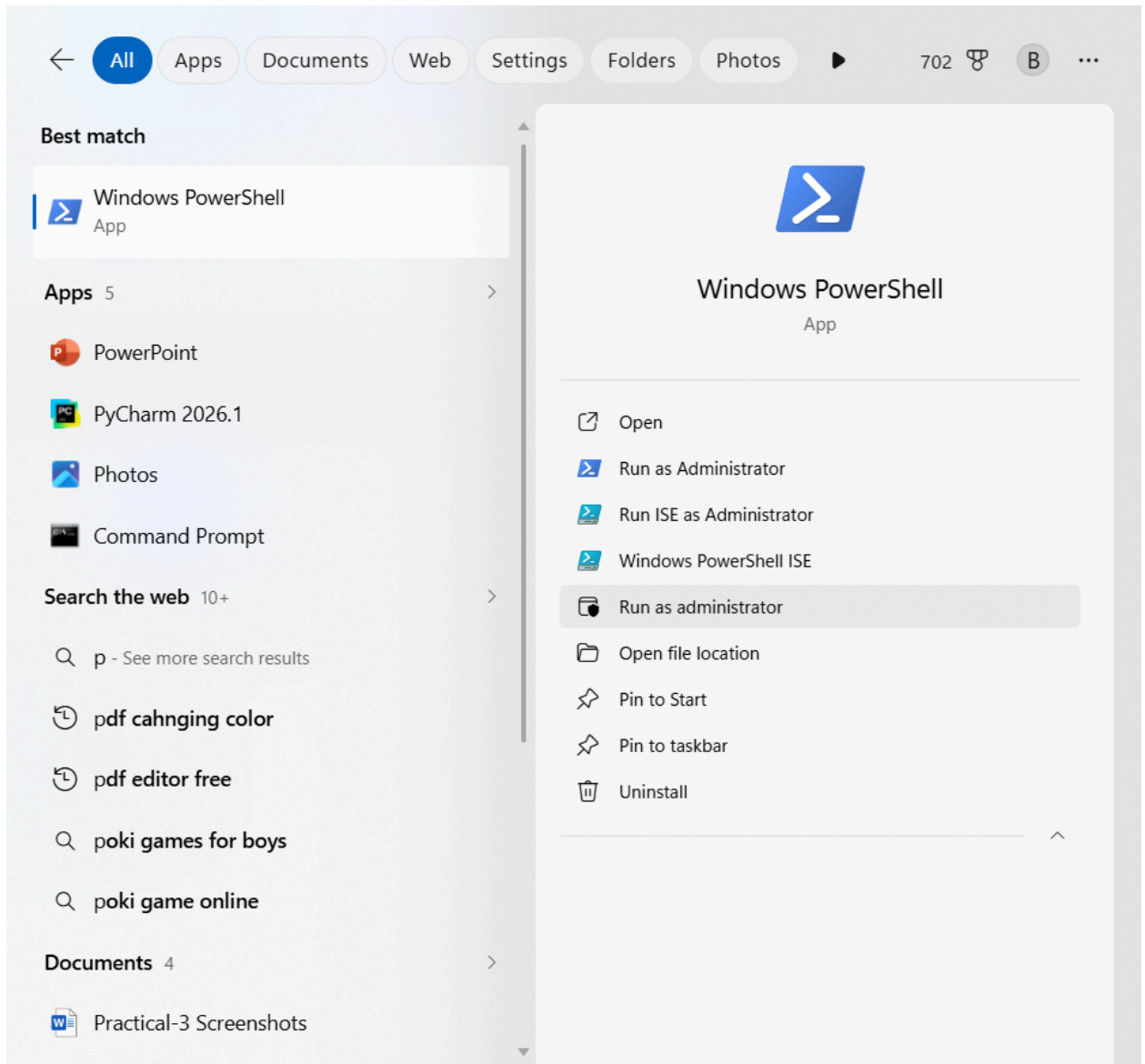


SKILL

TASK-1 :

1. Install/Verify Linux





1. Enter your username
2. Create a password

3. Confirm the password
4. Wait for Ubuntu to finish setup

```
bindhu@LAPTOP-NQPCRIKO:~$ uname -a
Linux LAPTOP-NQPCRIKO 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
IC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
bindhu@LAPTOP-NQPCRIKO:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 26.04 LTS
Release:        26.04
Codename:       resolute
bindhu@LAPTOP-NQPCRIKO:~$ pwd
/home/bindhu
```

2. Configure gcc

```
bindhu@LAPTOP-NQPCRIKO:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

bindhu@LAPTOP-NQPCRIKO:~$
```

3. Install gcc

```
bindhu@LAPTOP-NQPCRIKO:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages
[492 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packa
ges [243 kB]
Fetched 873 kB in 2s (428 kB/s)
62 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
bindhu@LAPTOP-NQPCRIKO:~$ sudo apt install gcc
gcc is already the newest version (4:15.2.0-5ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 62
bindhu@LAPTOP-NQPCRIKO:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

bindhu@LAPTOP-NQPCRIKO:~$
```

Create C program

4. Compile the program

```
bindhu@LAPTOP-NQPCRIKO:~$ nano prac1.c
bindhu@LAPTOP-NQPCRIKO:~$ gcc prac1.c
bindhu@LAPTOP-NQPCRIKO:~$ ./a.out
Parent Process ID: 1267
Parent's Parent Process ID: 735
```

5. Git repository

```
bindhu@LAPTOP-NQPCRIKO:~$ git --version
git version 2.53.0
```

6. Shell architecture

```
bindhu@LAPTOP-NQPCRIKO:~$ mkdir OSSP
bindhu@LAPTOP-NQPCRIKO:~$ cd OSSP
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warni
ng,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName fal
se"
Initialized empty Git repository in /home/bindhu/OSSP/.git/
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ |
```

7. Create project structure

bindhu@LAPTOP-NQPCRIKO: × + ▾

```
bindhu@LAPTOP-NQPCRIKO:~$ pwd
/home/bindhu
bindhu@LAPTOP-NQPCRIKO:~$ mkdir Practical
bindhu@LAPTOP-NQPCRIKO:~$ cd Practical
bindhu@LAPTOP-NQPCRIKO:~/Practical$ mkdir Skill-01
bindhu@LAPTOP-NQPCRIKO:~/Practical$ cd ..
bindhu@LAPTOP-NQPCRIKO:~$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
bindhu@LAPTOP-NQPCRIKO:~$ find .
.
./motd_shown
./open2.c
./exec3.c
./linux_TASK4
./task3
./landscape
./landscape/sysinfo.log
./fork.c
./prac1.c
./execex1.c*
./pipe2.c
./pipe2
./f3
./openb1.c
./b1
./a8
./open1.c
./exec1.c
./f1.c
./exec2.c
./ope2.c
./task2.c
./c2
./a7
./exechole.c
./pipe3.c
./read1.c
./o
```

```

bindhu@LAPTOP-NQPCRIKO: × + ▾
./.config/ubuntu-insights/ubuntu_release_upgrader-consent.toml
./.config/ubuntu-insights/linux-consent.toml
./.config/ubuntu-insights/wsl_setup-consent.toml
./.config/procps
./dir9
./.cache
./.cache/ubuntu-insights
./.cache/ubuntu-insights/wsl_setup
./.cache/ubuntu-insights/wsl_setup/uploaded
./.cache/ubuntu-insights/wsl_setup/uploaded/1784889516.json
./.cache/ubuntu-insights/wsl_setup/local
./.cache/ubuntu-insights/linux
./.cache/ubuntu-insights/linux/uploaded
./.cache/ubuntu-insights/linux/uploaded/1784889848.json
./.cache/ubuntu-insights/linux/local
./.cache/ubuntu-pro
./.cache/ubuntu-pro/ubuntu-pro.log
./.cache/motd.legal-displayed
./x1
./open2.cX
./student_portal
./student_portal/logs
./student_portal/backup
./student_portal/docs
./student_portal/config
./student_portal/src
./dir2
./newfile1
./readwrite.c
./pipe6.c
./b2
./pipe7.c
./dir1
./dir1/f3
./dir1/f2
./dir1/f1
./pipeex1.c
./sample.txt
./script.txt
./task1
bindhu@LAPTOP-NQPCRIKO:~$ |

```

8. Build Initial Makefile

```

bindhu@LAPTOP-NQPCRIKO:~$ cd OSSP
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ ls
Practical
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ |

```

9. Process abstraction

```
bindhu@LAPTOP-NQPCRIKO: ~$ echo $SHELL
/bin/bash
bindhu@LAPTOP-NQPCRIKO:~$ ps
  PID TTY          TIME CMD
  735 pts/2        00:00:01 bash
 1714 pts/2        00:00:00 ps
bindhu@LAPTOP-NQPCRIKO:~$ |
```

10. Create `fork()`

```
bindhu@LAPTOP-NQPCRIKO:~$ cd ~/OSSP
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ nano fork.c
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ gcc fork.c -o fork
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ ./fork
Before fork
After fork
After fork
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ |
```

11. `exec()` family

```
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ nano parent_child.c
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ gcc parent_child.c -o parent_child
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ ./parent_child
Parent Process
Child Process
Parent PID: 1887
Child PID: 1888
Parent PID: 1887
```

12. Parent-child relationship

```
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ nano exec.c
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ gcc exec.c -o exec
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ ./exec
Before exec
Practical  exec  exec.c  fork  fork.c  parent_child  parent_child.c
bindhu@LAPTOP-NQPCRIKO:~/OSSP$
```

13. Process tree


```

bindhu@LAPTOP-NQPCRIKO: X + v
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=4080, ...})
= 0
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/ls", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/ls/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
brk(0x57db3f358000) = 0x57db3f358000
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7b28be445cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7b28be445cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTHIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7b28be445cb0}, {sa_handler=0x57db2ee440c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7b28be445cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTHIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7b28be445cb0}, {sa_handler=0x57db2ee440c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7b28be445cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NLE|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x57db3f379000) = 0x57db3f379000
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NLE|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=19, ws_col=76, ws_xpixel=0, ws_ypixel=0}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=19, ws_col=76, ws_xpixel=0, ws_ypixel=0}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NLE|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NLE|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
statx(AT_FDCWD, "", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "", O_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
newfstatat(AT_FDCWD, "", {st_mode=S_IFDIR|0755, st_size=4096, ...}, 0) = 0
getdents64(3, 0x57db3f34a150 /* 10 entries */, 32768) = 288
getdents64(3, 0x57db3f34a150 /* 0 entries */, 32768) = 0
close(3) = 0
write(1, "Practical exec\t exec.c\t fork f"... , 68Practical exec      exec.c      fork fork.c parent_child parent_child.c
) = 68
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7b28be912000, 12288) = 0
exit_group(0) = 7
+++ exited with 0 +++
bindhu@LAPTOP-NQPCRIKO:~/OSSP$ |

```